

Course Code	Course Title				Course Type	
CS3102	Cloud Computing				PCC	
Prerequisite	Contact Hours per Week			Internal	External	Credits
	L	T	P			
	3	0	0	40	60	3

Course Objectives

- This course provides an insight into cloud computing.
- Topics covered include- distributed system models, different cloud service models, service oriented architectures, cloud programming and software environments, resource management.

Course Outcomes

1. Ability to understand various service delivery models of a cloud computing architecture.
2. Ability to understand the ways in which the cloud can be programmed and deployed.
3. Understanding cloud service providers.

Detailed Contents

UNIT-I

Computing Paradigms: High-Performance Computing, Parallel Computing, Distributed Computing, Cluster Computing, Grid Computing, Cloud Computing, Bio computing, Mobile Computing, Quantum Computing, Optical Computing, Nano computing.

UNIT-II

Cloud Computing Fundamentals: Motivation for Cloud Computing, The Need for Cloud Computing, Defining Cloud Computing, Definition of Cloud computing, Cloud Computing Is a Service, Cloud Computing Is a Platform, Principles of Cloud computing, Five Essential Characteristics, Four Cloud Deployment Models

UNIT-III

Cloud Computing Architecture and Management: Cloud architecture, Layer, Anatomy of the Cloud, Network Connectivity in Cloud Computing, Applications, on the Cloud, Managing the Cloud, Managing the Cloud Infrastructure Managing the Cloud application, Migrating Application to Cloud, Phases of Cloud Migration Approaches for Cloud Migration.

UNIT-IV

Cloud Service Models: Infrastructure as a Service, Characteristics of IaaS. Suitability of IaaS, Pros and Cons of IaaS, Summary of IaaS Providers, Platform as a Service, Characteristics of PaaS, Suitability of PaaS, Pros and Cons of PaaS, Summary of PaaS Providers, Software as a Service, Characteristics of SaaS, Suitability of SaaS, Pros and Cons of SaaS, Summary of SaaS Providers, Other Cloud Service Models.

UNIT-V

Cloud Service Providers: EMC, EMC IT, Captiva Cloud Toolkit, Google, Cloud Platform, Cloud Storage, Google Cloud Connect, Google Cloud Print, Google App Engine, Amazon Web Services, Amazon Elastic Compute Cloud, Amazon Simple Storage Service, Amazon Simple Queue ,service, Microsoft, Windows Azure, Microsoft Assessment and Planning Toolkit, SharePoint, IBM, Cloud Models, IBM Smart Cloud, SAP Labs, SAP HANA Cloud Platform, Virtualization Services Provided by SAP, Sales force, Sales Cloud, Service Cloud: Knowledge as a Service, Rack space, VMware, Manjra soft, Aneka Platform.

Text Books

- K. Chandrasekharan, "Essentials of Cloud Computing," CRC Press, USA: CRC Press, 2014.

References

1. R. Buyya, J. Broberg, and A. M. Goscinski, "Cloud Computing: Principles and Paradigms," Wiley, USA: John Wiley & Sons, Inc., 2011.
2. K. Hwang, G. C. Fox, and J. J. Dongarra, "Distributed and Cloud Computing," 1st ed., Elsevier, USA: Morgan Kaufmann Publishers, 2012.
3. T. Mather, S. Kumaraswamy, and S. Latif, "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance," O'Reilly, USA: O'Reilly Media, Inc., 2011.



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A.Y. 2024 – 2025_Sem 1
Lecture Plan

Faculty Name: D Devyani

Designation: Guest faculty

Subject Name: Cloud Computing

Subject Code: CS3102

Year:2024-2025

Department: CSE

Lecture No.	Unit No.	Topic Name	Reference Material Links	Reference Video Links (if any)
1	I	Definition of cloud computing, Computing Paradigm	R. Buyya, J. Broberg, and A. M. Goscinski, "Cloud Computing: Principles and Paradigms," Wiley, USA: John Wiley & Sons, Inc	https://youtu.be/dmGybCohHsw?si=4inMpghdXIc-8gcO
2	I	High-Performance Computing		
3	I	Parallel Computing		
4	I	Distributed Computing		
5	I	Cluster Computing		
6	I	Grid Computing		
7	I	Cloud Computing		
8	I	Bio computing		
9	I	Mobile Computing		
10	I	Quantum Computing		
11	I	Optical Computing,		

		Nano computing		
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12	II	Cloud Computing Fundamental		
13	II	Motivation for Cloud Computing		
14	II	The Need for Cloud Computing, Defining Cloud Computing		
15	II	Definition of Cloud computing		
16	II	Cloud Computing Is a Service		
17	II	Cloud Computing Is a Platform		
18	II	Principles of Cloud computing, Five Essential Characteristics		
19	III	Four Cloud Deployment Models		

20	III	Cloud Computing Architecture and Managemen		
21	III	Cloud architecture, Laye		
22	III	Anatomy of the Cloud, Network Connectivity in Cloud Computing		
23	III	Applications, on the Cloud, Managing the Cloud		
24	III	Managing the Cloud Infrastructure Managing the Cloud application		
25	III	Migrating Application to Cloud		
26	III	Phases of Cloud Migration Approaches for Cloud Migration		

27	IV	Cloud Service Models: Infrastructure as a Service		
28	IV	Characteristics of IaaS. Suitability of IaaS		
29	IV	Pros and Cons of IaaS, Summary of IaaS Providers		
30	IV	Platform as a Service, Characteristics of PaaS		
31	IV	Suitability of PaaS, Pros and Cons of PaaS		
32	IV	Summary of PaaS Providers, Software as a Service		
33	IV	Characteristics of		

Technologies

	IV	Summary of PaaS Providers, Software as a Service		
33	IV	Characteristics of SaaS		
34	IV	Suitability of SaaS,		
35	IV	Pros and Cons of SaaS, Summary of SaaS Providers		
36	IV	Other Cloud Service Models		

37	V	Cloud Service Providers: EMC, EMC IT, Captiva Cloud Toolkit		
38	V	Google, Cloud Platform, Cloud Storage		
39	V	Google Cloud Connect, Google Cloud Print, Google App Engine		
40	V	Amazon Web Services, Amazon Elastic Compute Cloud		
41	V	Amazon Simple Storage Service, Amazon Simple Queue		
42	V	service, Microsoft, Windows Azure		
43	V	Microsoft Assessment and Planning Toolkit, SharePoint, IBM, Cloud Models		
44	V	IBM Smart Cloud, SAP Labs, SAP HANA Cloud Platform		
45	V	Virtualization Services Provided by SAP, Sales force, Sales Cloud, Service Cloud		

Syllabus for MT-I: Lecture No. 1 to 15

Syllabus for MT-II: Lecture No. 16 to 36

Syllabus for MT-III: Lecture No. 37 to 45